

Custodial Symmetry in Randall-Sundrum

A pedagogical walkthrough

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Built up from the ground: where custodial symmetry comes from, how it is added to RS, what it modifies and why, the electroweak-versus-flavor split, and when the protection is valid. Each piece is tied to what our code actually does.

1 Why there is a problem at all

In Randall-Sundrum the SM gauge fields live in the 5D bulk between a UV brane (Planck scale) and an IR brane (TeV scale). Every 4D gauge boson is the *zero mode* of a 5D field, with an infinite tower of Kaluza-Klein (KK) excitations above it at the TeV scale. Electroweak symmetry breaking happens on (or near) the IR brane, where the Higgs lives. When the Higgs gets its vev it does not only give mass to the zero-mode W and Z , it also *mixes* them with their heavy KK partners; integrating those out shifts the properties of the light W and Z . Those shifts are what the LEP/SLC precision data measure, encoded in the oblique parameters S, T, U .

The dangerous one is T . It measures the breaking of *custodial isospin*, which controls

$$\rho = \frac{M_W^2}{M_Z^2 \cos^2 \theta_W},$$

measured to be 1 to high precision. In minimal RS the correction to T is not just nonzero, it is **enhanced by the warp volume**,

$$\Delta T_{\min} \simeq + \frac{\pi v^2}{2 \cos^2 \theta_W M_{KK}^2} L, \quad L \equiv k\pi r_c \approx 35-37,$$

where L is the logarithm of the Planck-to-TeV hierarchy, the very thing RS is built to generate. It is large and cannot be made small, so T forces the first KK gauge bosons up to roughly 10 TeV. This is the “RS T problem”: a little-hierarchy problem in which precision electroweak data push the new physics back up to ~ 10 TeV. It happens because the bulk gauge symmetry of minimal RS is just the SM’s $SU(2)_L \times U(1)_Y$, and the IR brane where EWSB happens has no reason to respect custodial isospin, so nothing forbids a large, volume-enhanced T .

2 Where the cure comes from: custodial symmetry in the SM

Step back to the ordinary Standard Model. Write the Higgs doublet as a 2×2 matrix $\Sigma = (\tilde{H}, H)$. The Higgs *potential* is invariant under a larger global symmetry than one might expect: $SU(2)_L \times SU(2)_R$, acting as $\Sigma \rightarrow L \Sigma R^\dagger$. The vev $\Sigma \rightarrow v \mathbb{1}$ breaks this to the diagonal subgroup $SU(2)_{L+R} \equiv SU(2)_V$, the **custodial symmetry**.

That diagonal symmetry is what guarantees $\rho = 1$ at tree level in the SM: the three would-be Goldstones (the longitudinal W, Z) form a custodial triplet, locking the W and Z masses in the symmetric ratio. T stays zero until something breaks $SU(2)_V$; in the SM only the hypercharge coupling g' and the top-bottom Yukawa splitting do, and both are small. The point: **custodial $SU(2)_V$ forbids T** . If you could impose it on the new physics, T would be protected.

3 How custodial is added to RS

The idea (Agashe, Delgado, May, Sundrum 2003, “ADMS”) is to *gauge* this custodial symmetry in the bulk, enlarging the bulk gauge group:

$$\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y \longrightarrow \mathrm{SU}(2)_L \times \mathrm{SU}(2)_R \times \mathrm{U}(1)_X.$$

Hypercharge is now a combination, $Y = T_R^3 + X$, with the extra $\mathrm{U}(1)_X$ essentially $(B - L)/2$. To recover the SM at low energy the extra symmetry is broken back with **boundary conditions**:

- On the **UV brane**, $\mathrm{SU}(2)_R \times \mathrm{U}(1)_X \rightarrow \mathrm{U}(1)_Y$ (Dirichlet BCs for the broken generators remove their zero modes and give the W_R, Z' KK modes UV-brane masses, so there is no light right-handed W or extra Z).
- On the **IR brane**, the Higgs is a *bidoublet* $(2, 2)$ of $\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$, and its vev breaks $\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R \rightarrow \mathrm{SU}(2)_V$, exactly as in the SM.

Now the IR brane, where EWSB happens, *respects the diagonal custodial* $\mathrm{SU}(2)_V$, so the large volume-enhanced T is forbidden by the bulk symmetry. The only custodial breaking left is the $\mathrm{SU}(2)_R \rightarrow \mathrm{U}(1)_Y$ breaking on the *UV* brane, far from the IR-localized fields, so the zero modes barely feel it.

4 What this does to T (and why S is untouched)

The result is the closed form our code uses, CGHNP Eq. 153:

$$\Delta T_{\mathrm{cust}} \simeq -\frac{\pi v^2}{4 \cos^2 \theta_W M_{KK}^2} \frac{1}{L}.$$

Compared with the minimal case the $+L$ became $-1/L$: the sign flipped and the volume *enhancement* became a volume *suppression*. The ratio is

$$\left| \frac{T_{\min}}{T_{\mathrm{cust}}} \right| = 2L^2 \approx 2450,$$

so custodial T is $\sim 2450\times$ smaller. The leading L -enhanced piece is now symmetry-forbidden; what survives is a small residual from the UV-brane breaking that the IR-localized zero mode only feels through its exponentially small UV tail, hence the $1/L$. In our run this pulls the oblique floor from ~ 16 to ~ 5.8 TeV.

The floor does not go to zero because **S is not protected**. S measures a different combination (roughly the W - B kinetic mixing from the KK tower), and custodial $\mathrm{SU}(2)_V$ says nothing about it. So once T is killed, S sets the residual floor (~ 2 – 3 TeV in the literature, ~ 5.8 TeV in our more conservative proxy). Custodial fixes T ; S survives and sets the wall.

5 The second problem, and the second tool: P_{LR}

Gauging $\mathrm{SU}(2)_R$ fixes the oblique T but not the $Z \rightarrow b\bar{b}$ vertex. The top is heavy, so the left-handed doublet $(t, b)_L$ is IR-localized ($c_{Q_3} < 1/2$); b_L lives in that doublet, so it is IR-localized too,

and IR-localized fermions overlap strongly with the KK gauge bosons and the Higgs. The $Zb_L\bar{b}_L$ coupling therefore gets a large correction δg_L^b , shifting R_b, A_b, A_{FB}^b and giving a bound independent of T .

The fix needs an additional *discrete* symmetry, P_{LR} , the exchange $L \leftrightarrow R$ inside the custodial group (Agashe, Contino, Da Rold, Pomarol 2006, “ACDP”). The dangerous correction comes from the broken generators and is *odd under* P_{LR} , so if b_L is a P_{LR} eigenstate the correction must cancel. P_{LR} exchanges $T_L^3 \leftrightarrow T_R^3$, so the protected point is

$$T_L^3 = T_R^3.$$

One arranges this by embedding $(t, b)_L$ in a **bidoublet** $(2, 2)_{2/3}$ where b_L sits at $(T_L^3, T_R^3) = (-\frac{1}{2}, -\frac{1}{2})$; there P_{LR} leaves it invariant and $\delta g_L^b \rightarrow 0$ at tree level. The price is physical: a bidoublet has four components, so besides $(t, b)_L$ it contains exotic partners, including a charge-5/3 quark and an extra charge-2/3 quark. These are the **custodians**, new TeV-scale KK fermions, and they are where the loop corrections below come from.

For the right-handed b : a singlet $(1, 1)_{-1/3}$ has $T_L^3 = T_R^3 = 0$, trivially P_{LR} -symmetric, so $\delta g_R^b = 0$. (Other embeddings, e.g. the $(2, 4)$ -type used to *fit* A_{FB}^b , break this and give a nonzero δg_R^b . So $g_R^b = 0$ is an embedding choice, not a theorem.)

6 The decomposition in our code

Custodial is therefore two protections doing two jobs, which is the `custodial_rs_su2r` vs `custodial_rs_plr` split in the code:

- $SU(2)_R$ protects the oblique T : floor $16 \rightarrow 5.8$ TeV. It does nothing to Zb_L .
- P_{LR} protects Zb_L : $\delta g_L^b \rightarrow 0$. It does nothing to the oblique floor.

Run side by side, `custodial_rs_su2r` and `custodial_rs_plr` give the *same* 5.8 TeV oblique floor but a *different* δg_L^b (the first leaves it at the minimal 2.6×10^{-3} , the second zeroes it). Two symmetries, two observables, independent.

7 Electroweak versus flavor

Custodial is an **electroweak** symmetry: it gauges $SU(2)_R \times U(1)_X$ and a discrete parity inside it, acting on electroweak quantum numbers. So the things it protects are electroweak: the oblique parameters and the Z couplings.

The RS **flavor** problem comes from somewhere else: the **KK gluon**. Anarchic 5D Yukawas plus geometric localization generate flavor-violating quark couplings to the first KK gluon, and KK-gluon exchange produces $\Delta F = 2$ operators; the one that dominates ε_K is the left-right operator C_4 , QCD- and chirally enhanced. Crucially, **the KK gluon is a color octet and an electroweak singlet**. It is neutral under $SU(2)_L, SU(2)_R, U(1)_X$, so the custodial group cannot act on it. Custodializing the model leaves the leading ε_K amplitude *exactly unchanged*.

This is why the electroweak floors drop to a few TeV but the ε_K wall sits at tens of TeV. It is physics, not a code limitation: Blanke et al. computed the entire $\Delta F = 2$ set inside the full

custodial model and still found $M_{KK} \gtrsim 18\text{--}30$ TeV, and Bauer II states C_4 “is a pure QCD effect ... insensitive to the precise embedding of the electroweak gauge symmetry.” What moves the ε_K wall is **flavor alignment** (RH-down $U(3)$ degeneracy, V_{5KM} , 5D-MFV), a different axis. Custodial and alignment are orthogonal cures for orthogonal problems: custodial for electroweak, alignment for flavor.

8 Subtleties: when “nothing” is not literally nothing

The same custodial structure that protects the diagonal Zb_L vertex also reshuffles the flavor-violating Z couplings. Bauer II Fig. 13 / Eq. 131 is exactly this: the left-handed Z -mediated FCNCs are suppressed by L , but the right-handed ones are *enhanced* by about a factor of 10. So custodial genuinely changes the tree-level Z contribution to $\Delta F = 2$. For ε_K this stays subleading to the KK gluon, so it does not move the wall; where it matters is rare decays like $K \rightarrow \pi\nu\bar{\nu}$. Hence the careful statement is “custodial does not move the ε_K wall,” not “custodial does literally nothing to flavor.”

9 When is the protection valid, and what are its limits

- **Tree-level, loop-corrected.** P_{LR} zeroes δg_L^b at tree level, but the custodian top-partners in loops (Carena, Pontón, Santiago, Wagner, “CPSW”) bring back a negative δg_L^b correlated with $+\Delta T$.
- **Embedding-dependent.** The protection holds only for $q_L \in (2, 2)_{2/3}$ and $b_R \in (1, 1)_{-1/3}$; other choices lose it (sometimes on purpose, to fit A_{FB}^b).
- **Leading-order in $1/L$.** The custodial T (Eq. 153) is the leading $1/L$ piece; there are subleading corrections.
- **It costs new states.** The bidoublets force the custodian fermions, including the charge-5/3 quark, into the TeV-scale spectrum.

For *our* implementation specifically: we do not build the full 5D custodial model, we impose its leading results as a proxy: swap in the custodial T coefficient, zero δg_L^b on the protected mask, zero δg_R^b for the singlet, add the CPSW loops, and leave the KK-gluon $\Delta F = 2$ untouched. The four places where that proxy is softer than a real model (the κ_b/L residual, the ξ/ρ loop-mass inputs, the unpinned g_R embedding, the singlet-only ΔT) are documented in `docs/CUSTODIAL_PROVENANCE.md` and are what a full custodial build (roadmap D) would replace with derived numbers.

In four sentences. Minimal RS has a volume-enhanced T problem and a Zb_L problem, both electroweak. Gauging a bulk custodial $SU(2)_R$ makes the IR brane respect custodial isospin, killing the enhanced T and leaving the unprotected S as the residual floor. Adding the discrete P_{LR} with the bidoublet embedding protects Zb_L on top of that. None of it touches ε_K , set by the color-octet, electroweak-singlet KK gluon, so flavor needs its own separate cure: alignment.